Vo Van Tu Tai

Full-Stack Software Engineer (Node.js · PHP · React · PostgreSQL)

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PROFESSIONAL SUMMARY

Information Systems Engineer (IUH, GPA 3.26/4.0 - Distinction) with proven hands-on full-stack development experience across **Node.js (Express, NestJS), PHP (OOP & MVC), JavaScript/TypeScript, and React**. Proficient in relational data modeling (**PostgreSQL, MySQL 3NF**), query performance tuning (reduced latency by 40%), and concurrency control (ACID transactions, pessimistic locking). Author of a live logistics order tracking & AI platform operating at **tracking.nexus-ex.site**. Strong capability in reading English technical specifications; adept at AI-assisted engineering (Cursor, Claude) to accelerate delivery. Available immediately for full-time employment.

CORE TECHNICAL SKILLS

- **Backend & API Engineering:** Node.js (Express.js, NestJS), PHP (OOP, Custom MVC Framework), RESTful APIs (OpenAPI 3.0 specs), Webhooks, Message Brokers (RabbitMQ), Concurrency Control (ACID Transactions, Locking).
- **Frontend & User Interface:** JavaScript (ES6+), TypeScript, React.js, HTML5, Modern CSS, Responsive Web Design, DOM Manipulation, Single Page Applications (SPA).
- **Databases & Caching:** PostgreSQL, MySQL (Normalized 3NF, B-Tree Indexing, ACID Compliance), Redis (In-memory Caching, Distributed Locks).
- **DevOps & Engineering Practices:** SDLC Lifecycle, Agile/Scrum, Git/GitHub (Branch management, pull requests, conflict resolution), Docker, Docker Compose, Linux/Bash basics.
- **AI Tooling & Languages:** Cursor, Claude, ChatGPT, GitHub Copilot (rapid prototyping, clean code & testing); Technical English (Reading engineering docs, API specifications, and basic communication), Vietnamese (Native).

WORK EXPERIENCE

HomeNest Software
Ho Chi Minh City, Vietnam

Technical Bridge & Full-Stack Engineer (Technical Bridge · BA / PM Assistant)
Mar 2026 – Sep 2026

Project 1: "Life Assistance" — Multi-Service On-Demand Platform & Dynamic Bidding

- **Multi-Service Architecture & API Contracts:** Analyzed and modeled data architectures for an on-demand ecosystem (Handyman, Ride-Hailing, Carpooling, Super Admin); authored standardized JSON contracts and OpenAPI 3.0 specifications for backend engineering teams.
- **Dynamic Bidding State-Machine Design:** Designed two-way negotiation state-machines (passenger quote ↔ driver counter-bid) featuring TTL expirations and distributed state locking, eliminating race conditions during transaction settlement.
- **Event-Driven GPS Telemetry Optimization:** Proposed event-triggered GPS coordinate logging (strictly upon ride pickup and drop-off milestones rather than continuous polling), cutting server network overhead, preserving device battery, and reducing third-party map API expenses.
- **Technical Specifications & Relational Schema Review:** Authored comprehensive PRDs and User Stories with testable acceptance criteria (Given-When-Then); collaborated with Tech Lead to review Figma wireframes against 3NF relational database models, eliminating schema discrepancies prior to sprint kickoffs.

Project 2: "VietPlus" — Classified Marketplace & Cross-Border Logistics

- **Domain Mastery & Listing Lifecycle:** Rapidly mastered system backlog and C2C architecture; engineered listing lifecycles, automated content moderation rules, and streamlined data pipelines to maintain steady sprint velocity.
- **Fulfillment, Order Cancellation & Returns State-Machine:** Directly gathered and structured client Change Requests; modeled comprehensive state-machines governing parcel delivery, automated timeout cancellations, and refund dispute escalation workflows.
- **Pragmatic Scope Governance (Lead Form Intake):** Advised stakeholders to deploy a lightweight Lead Intake Form workflow for urgent cross-border shipping needs instead of building an end-to-end logistics subsystem, successfully safeguarding the sprint's Scope Freeze.
- **Functional QA Testing & UAT Sign-off:** Formulated mobile app release verification checklists (App Store, Google Play); conducted functional and smoke testing on staging environments based on User Stories, managed defect triage in Jira, and facilitated final UAT client sign-off.

WORK EXPERIENCE (CONTINUED)

J&T Express Ho Chi Minh City, Vietnam

Operations Process Coordinator & Data Analyst Sep 2025 – Feb 2026

- **SQL Data Auditing & Integrity:** Formulated daily SQL reconciliation queries (JOIN, Aggregation) cross-checking physical manifests against logistics databases; ensured supply-chain data consistency across high-volume shifts.
- **One-Way Operational Throughput:** Analyzed transit bottlenecks and restructured parcel sorting into a closed one-way workflow (Inbound → Weigh → Scan → Dispatch), clearing hub congestion and boosting throughput.
- **Standard Operating Procedures (SOP):** Authored operational SOPs classifying shipment anomalies; guided hub personnel in precise system execution, minimizing human data entry errors.

REPRESENTATIVE TECHNICAL PROJECTS

Nexus Express — Order Management System (OMS) & Live AI Tracking [Live Demo ↗] [GitHub ↗] Distributed Architecture Project

NestJS, TypeScript, PostgreSQL (3NF), RabbitMQ, React, Live RAG, Docker Compose Sep 2025 – Feb 2026

- **Event-Driven Distributed Architecture:** Engineered an enterprise order management platform live at tracking.nexus-ex.site; leveraged NestJS and RabbitMQ message queues to decouple dispatch and parcel tracking streams, absorbing surge traffic without bottlenecks.
- **Idempotent API Contracts & BPMN Workflows:** Designed OpenAPI 3.0 contracts mandating Idempotency Keys to prevent duplicate processing during network retries; modeled parcel lifecycles using Camunda BPMN 2.0 to enforce strict status gateways.
- **PostgreSQL 3NF Modeling & Audit Logs:** Architected a normalized 3NF schema with composite B-Tree indexes on tracking codes and timestamps; maintained tamper-proof event audit logs enabling 100% transit traceability.
- **Live RAG Assistant Integration:** Integrated a live retrieval-augmented generation (RAG) assistant leveraging tool/function calling connected directly to backend services, enabling automated parcel tracking and instant rate quotation.

Song Tai Shop & Tourist Accommodation — E-Commerce & Booking Platform [GitHub ↗] Web Application Project

PHP (OOP & Custom MVC), MySQL (3NF & ACID), VNPay IPN Webhook, Concurrency Control, Docker Oct 2025 – Jan 2026

- **Custom MVC Framework & Security:** Developed a custom PHP MVC framework featuring Front-Controller (index.php), REST-style regex routing, PDO prepared statements preventing SQL injections, CSRF tokens, and session-based RBAC.
- **ACID Concurrency Control & TTL Holds:** Prevented inventory overselling and overbooking race conditions using strict MySQL ACID transactions (START TRANSACTION, COMMIT) coupled with pessimistic row locking (SELECT ... FOR UPDATE) and 15-minute temporary holds.
- **Encrypted Webhook Integration & Optimization:** Integrated VNPay payments using HMAC-SHA512 checksum validation with idempotent webhook handlers; optimized 3NF MySQL schemas with composite indexes achieving sub-second query response times.

Cab Booking & Real-Time Dispatch System — High-Concurrency Backend [GitHub ↗] High-Concurrency System Project

Node.js, Express.js, Redis (In-Memory Caching & Distributed Locks SETNX), Leaflet, Docker Dec 2025 – Jan 2026

- **High-Concurrency Matching via Redis Distributed Locks:** Built a driver-passenger dispatch engine using Redis distributed locks (SETNX with TTL), eliminating race conditions when multiple drivers simultaneously accept a ride request.
- **In-Memory Caching & Geospatial Queries:** Leveraged Redis geospatial indexes (GEOADD, GEORADIUS) to track real-time driver coordinates and compute proximity in sub-milliseconds, bypassing disk queries on primary databases.
- **Trip State-Machine Governance:** Modeled strict ride state lifecycles (Requested → Accepted → Arrived → In-Trip → Completed/Cancelled) with dynamic surge pricing logic; packaged microservices into Docker Compose with health checks.

EDUCATION & ACADEMIC BACKGROUND

Industrial University of Ho Chi Minh City (IUH) Ho Chi Minh City, Vietnam

Engineer in Information Systems (Completed all degree requirements · Available Full-Time) Sep 2022 – 2026

- **Cumulative GPA: 3.26 / 4.0** (Distinction / Xếp loại Giỏi).
- **Relevant Coursework:** Software Engineering, Object-Oriented Programming (OOP), Data Structures & Algorithms, Systems Analysis & Design (OOAD/UML), Advanced Database Systems & SQL (MySQL, PostgreSQL, Indexing, Transactions), SOA, ERP Systems (Odoo).